BMS PROJECT 1

Systematic Review: Modeling Approaches for Antibody Afucosylation Mechanistic · Machine Learning · Hybrid Models · AI

Presented By:

TEAM DATA MINDS

Sanjith Ganesh (sg2151)

Pranav Senthilkumaran (ps1471)

Problem Statement: What Are We Solving?

Afucosylation = absence of core fucose on N-linked glycans → Critical Quality Attribute (CQA) in mAb biologics manufacturing → Directly impacts ADCC efficacy, patient safety, and regulatory approval.

The Challenge:

- Current prediction methods are **slow**, costly, or lab-scale only.
- No scalable industrial solution exists.
- **50+ process variables** interact in complex, nonlinear ways.

The Opportunity — ML & AI:

- Predict afucosylation in real time or early in the process using routinely measured process variables.
- Replace slow analytical assays with fast, data-driven surrogate models.
- Enable proactive batch correction before CQA goes out of specification.
- Build interpretable models that satisfy regulatory (FDA Quality by Design) requirements.
- Generalize across multiple mAb products using a single unified modeling framework.

Our Targeted Approach:
Systematic literature review + + **7 ML/AI models benchmarked** on academic and BMS industrial datasets — evaluated on accuracy, interpretability, data requirements, and regulatory usability.

Core Literature: 3 Reference Papers

Feature	Paper 1 · Antonakoudis 2021	Paper 2 · Kotidis 2019	Paper 3 · Kotidis 2020
Model Type	Hybrid (FBA + ANN)	Mechanistic (Kinetic)	ANN
Inputs	Extracellular metabolites	Intracellular NSDs + kinetics	Intracellular NSDs
Outputs	Glycoform distribution	Glycoform profile over time	18 glycan species
Key Idea	FBA fluxes → ANN	Golgi as plug-flow reactor	Shallow ANN, sigmoid activation
Accuracy	Good, updatable	Good within range	~1.1% avg abs. error
Interpretability	Medium	High	Low (black-box)
Comp. Cost	Low–Medium	High	Low
Best For	In-process monitoring	Mechanistic insight	Fast surrogate

Additional Findings: 3 New Papers

	New Paper A · BMS 2023	New Paper B · Tsopanoglou 2021	New Paper C · Anti-HER2 2024
Model	PLSR + VIP scores	Review paper	RF + PLSR
Focus	Afucosylation drivers (direct)	Hybrid modeling strategies	Glycan from media markers
Data	Industrial — 4 IgG1 mAbs, BMS	N/A	CHO-K1, 12 conditions
Key Insight	PLSR identifies key process levers	Hybrid models = future of Biopharma 4.0	Media variables predict glycan quality
Relevance	Replicates our statistical baseline; BMS-sponsored	Frames our hybrid model rationale	Validates media-driven ML approach

Full Comparison: All 6 Papers

Feature	P1 Antonakoudis	P2 Kotidis 2019	P3 Kotidis 2020	New A BMS 2023	New B Review	New C ML 2024
Model Type	Hybrid	Mechanistic	ANN	PLSR	Review	RF / PLSR
Afucosylation?	Indirect	Indirect	Yes (included)	Direct	General	Yes
Data Source	Lab (Lonza)	Lab (CHO)	Lab (4 proteins)	Industrial (BMS)	N/A	Lab (CHO-K1)
Open Data?	No	Partial	Yes	No	N/A	Partial
Practical Use	In-process control	Process design	Fast surrogate	Retrospective analysis	Strategy selection	Media optimization

Dataset Overview

Dataset 1 — Kotidis 2020 (Academic)

- **Source:** Imperial College London
- 132 data points · Days 7, 9, 11, 12 **(small dataset)**
- **12 inputs:** 6 nucleotides (AMP, ADP, ATP, CTP, UTP, GTP) + 6 NSDs (UDP-Gal, UDP-GlcNAc, GDP-Fuc...)
- **Outputs:** up to 18 glycoform species · afucosylation derived
- Open access (PMC)

Dataset 2 — BMS 2023 (Industrial) ← Primary Dataset

- **Source:** Bristol Myers Squibb
- 4 IgG1 mAb products · hundreds of manufacturing batch records
- **50+ inputs:** temperature, pH, DO, pCO2, osmolality, amino acids, vitamins, trace metals, Mn, uridine, galactose, VCD, viability
- **Outputs:** complex afucosylation %, high mannose afucosylation %, total afucosylation %
- Proprietary — to be obtained from BMS

Feature	Dataset 1	Dataset 2
Scale	Lab	Manufacturing
Samples	132	Hundreds of batches
Features	12 (intracellular)	50+ (extracellular)
Afucosylation	Derived	Direct output
Access	Open	Proprietary (BMS)
Best Models	ANN, GPR, Hybrid	PLSR, RF, XGBoost

Algorithm Implementation: 7 ML Modeling Approaches

#	Model	Paradigm	Accuracy	Interpretability	Cost	Best For
1	**MLR / Ridge**	Statistical baseline	Low–Med	Very High	Very Low	Performance floor, regulatory docs
2	**PLSR**	Dim. reduction + regression	Medium	High (VIP scores)	Low	Reproduce BMS paper
3	**Random Forest**	Ensemble trees	High	Medium	Low–Med	Nonlinear relationships
4	**ANN**	Deep learning	High (~1.1% error)	Low	Medium	Reproduce Paper 3
5	**GPR**	Probabilistic / Bayesian	High + UQ	Medium	Medium	Uncertainty bounds, small data
6	**Hybrid FBA+ANN**	Physics-informed	High	Medium–High	Medium	In-process monitoring, Paper 1
7	**XGBoost + SHAP**	Boosted trees + XAI	Very High	High (SHAP)	Low–Med	Best accuracy + explainability

Model Selection Matrix

Criterion	MLR/Ridge	PLSR	RF	ANN	GPR	Hybrid	XGBoost
Accuracy	Low	Med	High	High	High	High	Very High
Interpretability	High	High	Med	Low	Med	Med	High
Handles Multicollinearity					Partial		
Uncertainty Quantification	Partial	Partial	Partial				
Biological Grounding							
Works on Dataset 1	Limited	Limited	Limited				Limited
Works on Dataset 2							
Regulatory Friendliness	High	High	Med	Low	Med	Med	Med
Replicates Paper	—	New Paper A	—	Paper 3	—	Paper 1	—

Intersection of AI Beyond Prediction

AI Capability	How It Works	Applied To
Automated Data Cleansing	Outlier detection via IsolationForest / Z-score · Missing value imputation via KNN Imputer · Drift flagging on column statistics	BMS Dataset 2 (50+ variables)
AI Feature Selection	SHAP values auto-computed from XGBoost / RF · Correlation heatmap · SelectKBest ranking	Pre-processing for all 7 models
LLM Report Generation	SHAP summary + model metrics fed into Claude / GPT API via structured prompt · Returns plain-English explanation of key afucosylation drivers	Post-modeling regulatory reporting

Final Deliverable — Interactive Dashboard

Feature	Description
Data Upload	Upload BMS or custom dataset
Auto Data Cleansing	**One-click outlier detection**, missing value imputation, drift flagging
Feature Selection	SHAP or correlation-based ranking of input variables
Model Selection	Dropdown to choose from all 7 models **(MLR, PLSR, RF, ANN, GPR, Hybrid, XGBoost)**
Results View	Predicted afucosylation % with **actual vs predicted plots** and **error metrics**
LLM Explainer	Auto-generated plain-English summary of model outputs and key drivers
Report Export	Download regulatory-style PDF summary